Book strip

Science · GROW · Autumn 2 · Week 2 · GROW_SCI_A2_W02_P1

Spherical bodies: introduce and explore · Weekly arc: Introduce and explore.

Goal. I can compare a disc and a sphere across viewpoints and explain spherical evidence.

Today I can. I can distinguish a flat disc from a three-dimensional sphere using three viewpoints. I can keep viewing distance and angle equal so the shape comparison is fair. I can use space evidence and model limits to describe bodies as approximately spherical.

Made by Matt

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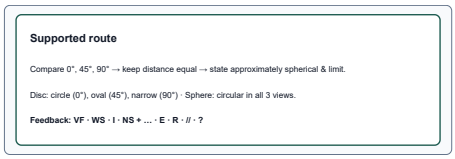
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Route card

Staff: cut these cards apart before the lesson. Give one route card. Mark actual **I · S · C** beside each task.

Supported



Supported route card. Compare disc and sphere viewpoints and explain spherical evidence.

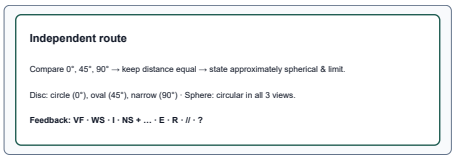
Use Reference A. Compare the flat disc and the solid sphere at 0° (front), 45° (oblique), and 90° (edge-on). Write or point to the outline for each view.

Use Reference A and Reference B. Work in your book or usual mode.

Staff I ☐ S ☐ C ☐ · Feedback: VF ☐ WS ☐ E ☐ NS + … ☐ R ☐ // ☐ ? ☐

My response:

Independent



Independent route card. Compare disc and sphere viewpoints and explain spherical evidence.

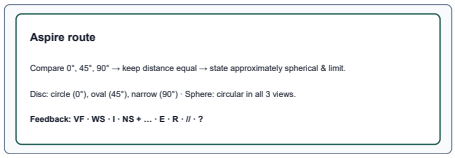
Use Reference A. Record the outlines of a flat disc and a sphere across three controlled viewpoints (0°, 45°, and 90°). State the two fairness controls.

Use Reference A and Reference B. Work in your book or usual mode.

Staff I ☐ S ☐ C ☐ · Feedback: VF ☐ WS ☐ E ☐ NS + … ☐ R ☐ // ☐ ? ☐

My response:

Aspire



Aspire route card. Compare disc and sphere viewpoints and explain spherical evidence.

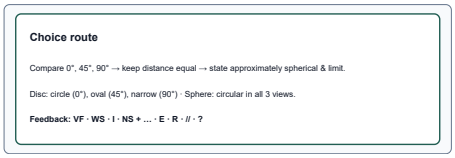
Use Reference A. Contrast the outlines of a flat disc and a sphere at 0°, 45°, and 90°. Keep viewing distance equal.

Use Reference A and Reference B. Work in your book or usual mode.

Staff I ☐ S ☐ C ☐ · Feedback: VF ☐ WS ☐ E ☐ NS + … ☐ R ☐ // ☐ ? ☐

My response:

Choice



Choice route card. Compare disc and sphere viewpoints and explain spherical evidence.

1. Look at Reference A. At 0° (front view), do the disc and sphere look the same (circular) or different? 2. At 90° (edge-on), which object looks narrow: the disc or the sphere?

Use Reference A and Reference B. Work in your book or usual mode.

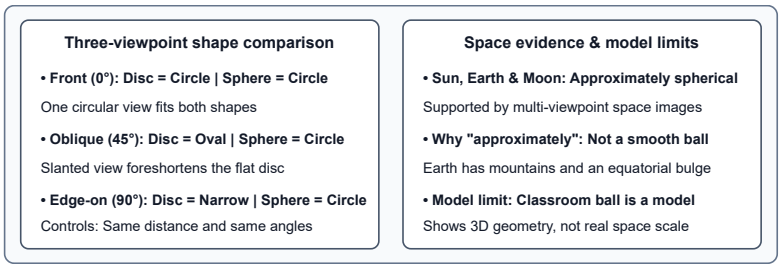
Staff I ☐ S ☐ C ☐ · Feedback: VF ☐ WS ☐ E ☐ NS + … ☐ R ☐ // ☐ ? ☐

My response:

Made by Matt

Reference sheets

Reference A · Three-viewpoint comparison: Flat disc vs Solid sphere

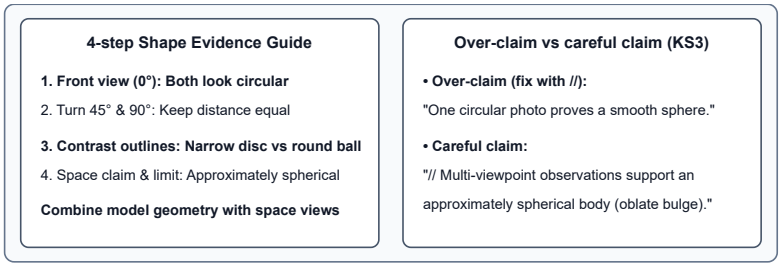


Three-viewpoint shape comparison. It compares a flat disc and a sphere at 0°, 45°, and 90°.

Reference only. Do not write on this sheet. Both objects are held at the same viewing distance across all three viewpoints.

Viewpoint angle.	Flat card disc outline.	Solid sphere (ball) outline.	What this viewpoint tells us.
1. Front view (0°).	Circular outline.	Circular outline.	Both shapes look circular from the front, so one view cannot separate them.
2. Oblique view (45°).	Oval (elliptical) outline.	Circular outline.	A slanted view foreshortens the flat disc while the sphere stays round.
3. Edge-on view (90° quarter-turn).	Narrow thin edge.	Circular outline.	Turning 90° separates the flat disc (narrow) from the 3D sphere (circular).

Reference B · Space evidence, careful claims, and model limits



Four-step Shape Evidence guide. It links 0°, 45°, and 90° views to spacecraft evidence and limits.

Evidence summary.	Key Stage 3 science meaning.	Honest model limit or careful wording.
E1 · Earth from spacecraft.	Spacecraft observations from multiple viewpoints show Earth as a three-dimensional round globe.	Earth has surface mountains and a small equatorial bulge, so it is approximately spherical.
E2 · Moon and Sun.	Multi-angle astronomical observations show the Moon and Sun have circular outlines from every viewpoint.	Described as approximately spherical three-dimensional bodies rather than flat discs.
E3 · Classroom ball model.	Predicts how a 3D sphere keeps a circular outline at 0°, 45°, and 90°.	Limit: the classroom ball is a geometric model, not real

Evidence summary.	Key Stage 3 science meaning.	Honest model limit or careful wording.
		planetary proof or to scale.

Made by Matt

Catch-up sheet

Absence or access only. This sheet gives a fast entry if a pupil joined late. Never add it on top of today's routine tasks.

1. **Three viewpoints (Reference A).** At **0° (front)** both a flat disc and a sphere look circular. At **45° (oblique)** the disc looks oval. At **90° (edge-on)** the disc looks narrow while the sphere stays circular.
2. **Fair comparison.** Keep viewing distance and viewing angles the same for both objects.
3. **Approximately spherical.** Space views show the Sun, Earth, and Moon are approximately spherical. Earth has mountains and a bulge. Our ball is a model.
4. **Quick link.** Point to Reference A and B. Then join today's You do Task 1.

Point, speak, sign, draw, type, write, or tell an adult. You may pass.

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Choice route sheet

Circle or point to your choice. Give a short reason from Reference A and Reference B.

1. 1. Look at Reference A. At 0° (front view), do the disc and sphere look the same (circular) or different? 2. At 90° (edge-on), which object looks narrow: the disc or the sphere? **Staff I ☐ S ☐ C ☐**

Help: Check Rows 1 and 3 on Reference A.

2. 3. Should viewing distance be the same or different for a fair test? 4. Can one front photo alone prove a sphere: No or Yes? **Staff I ☐ S ☐ C ☐**

Help: Check Row 1 and the fairness rule on Reference A.

3. 5. Look at Reference B. Which phrase best describes Earth, Sun, and Moon: approximately spherical or flat discs? 6. Name one limit of a smooth ball model: omits mountains/scale, or is real Earth?
Staff I ☐ S ☐ C ☐

Help: Check Reference B.

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Choice

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt